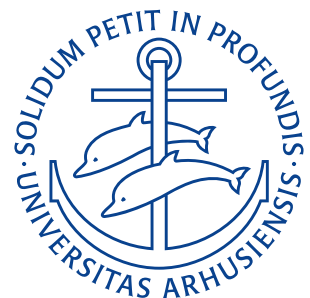


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Homework Exercises

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Lecture 1: Introduction

26. Januar 2026

Problem 1.1a

Show that

$$\delta_{ij}\delta_{ij} = 3.$$

DerivationExpanding $\delta_{ij}\delta_{ij}$ we get

$$\delta_{ij}\delta_{ij} = \delta_{11}\delta_{11} + \delta_{12}\delta_{12} + \delta_{13}\delta_{13} + \delta_{21}\delta_{21} + \delta_{22}\delta_{22} + \delta_{23}\delta_{23} + \delta_{31}\delta_{31} + \delta_{32}\delta_{32} + \delta_{33}\delta_{33}.$$

As $\delta_{11} = \delta_{22} = \delta_{33} = 1$ the above becomes:

$$\delta_{ij}\delta_{ij} = 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 = 3.$$

$\delta_{ij}\delta_{ij} = 3$

RESULT

Problem 1.1c

Show that

$$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}.$$

Derivation

We expand by summing over the common index j :

$$\alpha_{ij}b_{jk} = \sum_j \alpha_{ij}b_{jk} = \alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k}.$$

Now we sum over n instead:

$$\alpha_{i1}b_{1k} + \alpha_{i2}b_{2k} + \alpha_{i3}b_{3k} = \sum_n \alpha_{in}b_{nk}.$$

Q.E.D.

$\alpha_{ij}b_{jk} = \alpha_{in}b_{nk}$

RESULT

Problem 1.1e

Show that

$$\delta_{ij}C_{jkl} = C_{ikl}.$$

Derivation

Expanding we get

$$\delta_{ij}C_{jkl} = \delta_{11}C_{1kl} + \delta_{22}C_{2kl} + \delta_{33}C_{3kl} = C_{ikl}.$$

$\delta_{ij}C_{jkl} = C_{ikl}$

RESULT

Lecture 2: Traction, Stress, and Equilibrium

2. Februar 2026

Lecture 3: The Cauchy Theorem & Principal Stresses

9. Februar 2026

Problem 2.1

Find the components of the traction on a plane defined by $n_1 = 1/\sqrt{2}$, $n_2 = 1/\sqrt{2}$, and $n_3 = 0$ for the following states of stress:

- (a) $\sigma_{11} = \sigma, \sigma_{12} = 0, \sigma_{13} = 0, \sigma_{21} = \sigma, \sigma_{22} = 0, \sigma_{33} = \sigma.$
 (b) $\sigma_{11} = \sigma, \sigma_{12} = \sigma, \sigma_{13} = 0, \sigma_{22} = \sigma, \sigma_{23} = 0, \sigma_{33} = 0.$

Derivation

- (a) The stress state in (a) is

$$\sigma = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix}.$$

The Cauchy Theorem states that

$$T_i = \sigma_{ji} n_j \implies T = \sigma \cdot n.$$

Our plane is:

$$n = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

Hence,

$$T = \begin{bmatrix} \sigma & 0 & 0 \\ \sigma & 0 & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

- (b) For (b) our stress state is:

$$\sigma = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix},$$

so

$$T = \begin{bmatrix} \sigma & \sigma & 0 \\ \sigma_{21} & \sigma & 0 \\ \sigma_{31} & \sigma_{32} & 0 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix} = \begin{pmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{pmatrix}.$$

(a) $T = \begin{bmatrix} \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}$

(b) $T = \begin{bmatrix} \sqrt{2}\sigma \\ \frac{\sigma}{\sqrt{2}} + \frac{\sigma_{21}}{\sqrt{2}} \\ \frac{\sigma_{31}}{\sqrt{2}} + \frac{\sigma_{32}}{\sqrt{2}} \end{bmatrix}.$

RESULT

Problem 2.6

At a given point in a body,

$$\sigma_{11} = \sigma_{22} = \sigma_{33} = -p,$$

$$\sigma_{12} = \sigma_{23} = \sigma_{31} = 0.$$

Show that the normal stresses are equal to $-p$ and that the shearing stresses vanish for any other Cartesian coordinate system.

Derivation

Here, our stress tensor is

$$\boldsymbol{\sigma} = \begin{bmatrix} -p & 0 & 0 \\ 0 & -p & 0 \\ 0 & 0 & -p \end{bmatrix} = -p\mathbf{I}.$$

Using the Cauchy theorem $\mathbf{T} = \boldsymbol{\sigma} \cdot \mathbf{n}$ we get:

$$\mathbf{T} = (-p\mathbf{I}) \cdot \mathbf{n} = -p\mathbf{n}.$$

Hence, $\mathbf{T}$ is parallel to $\mathbf{n}$ meaning the traction does not have any components tangent to the plane and hence the stress is entirely normal with size $-p$.

Lecture 4: Equilibrium Equations & Principle of Angular Momentum

16. Februar 2026

Problem 2.2

The state of stress at a point P in a structure is given by

$$\sigma_{11} = 20000,$$

$$\sigma_{22} = -15000,$$

$$\sigma_{33} = 3000$$

$$\sigma_{12} = 2000,$$

$$\sigma_{23} = 2000,$$

$$\sigma_{31} = 1000.$$

- Compute the scalar components T_1 , T_2 , and T_3 of the traction $\mathbf{T}$ on the plane passing through P whose outward normal vector $\mathbf{n}$ makes equal angles with the coordinate axes x_1 , x_2 , x_3 .
- Compute the normal and tangential components of stress on this plane.
- Determine the principal stresses.
- Determine the direction of maximum principal stress.

Derivation

- We are given the following stress state:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & \cdot \\ \cdot & -15000 & 2000 \\ 1000 & \cdot & 3000 \end{bmatrix}.$$

We assume this is symmetric i.e. it satisfies the standard case of $\sigma_{ij} = \sigma_{ji}$ and we therefore get:

$$\boldsymbol{\sigma} = \begin{bmatrix} 20000 & 2000 & 1000 \\ 2000 & -15000 & 2000 \\ 1000 & 2000 & 3000 \end{bmatrix}.$$

As $\mathbf{n}$ makes equal angles with x_1 , x_2 , and x_3 , $\mathbf{n}$ must be:

$$\mathbf{n} = \frac{1}{\sqrt{3}}(1, 1, 1).$$

The traction vector on the plane is:

$$\mathbf{T} = \boldsymbol{\sigma} \mathbf{n}$$

so,

$$\begin{aligned} T_1 &= \frac{1}{\sqrt{3}} (20000 + 2000 + 1000) = \frac{23000}{\sqrt{3}} \\ T_2 &= \frac{1}{\sqrt{3}} (2000 - 15000 + 2000) = -\frac{11000}{\sqrt{3}} \\ T_3 &= \frac{1}{\sqrt{3}} (1000 + 2000 + 3000) = \frac{6000}{\sqrt{3}}. \end{aligned}$$

(b) The normal stress is given as

$$\sigma_{nn} = \mathbf{T}_n \cdot \mathbf{n}.$$

I.e.

$$\begin{aligned} \sigma_{nn} &= \frac{1}{\sqrt{3}} (T_1 + T_2 + T_3) \\ &= \frac{1}{3} (23000 - 11000 + 6000) \\ &= 6000. \end{aligned}$$

Hence, the normal traction vector $\mathbf{N}$ is

$$\mathbf{N} = \sigma_{nn} \cdot \mathbf{n} = 6000 \cdot \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

As the total traction $\mathbf{T}$ is the algebraic sum of the normal traction $\mathbf{N}$ and the tangential traction $\mathbf{S}$ we get:

$$\mathbf{T} = \mathbf{N} + \mathbf{S} \implies \mathbf{S} = \mathbf{T} - \mathbf{N}.$$

So

$$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix} - \frac{6000}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}.$$

(d) For principal stresses we have the condition:

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0$$

i.e.:

$$\begin{vmatrix} 20000 - \lambda & 2000 & 1000 \\ 2000 & -15000 - \lambda & 2000 \\ 1000 & 2000 & 3000 - \lambda \end{vmatrix} = 0 \rightarrow \lambda_1 \rightarrow -15319,7, \quad \lambda_2 \rightarrow 3133,4, \quad \lambda_3 \rightarrow 20186,3.$$

Therefore our principal stresses are:

$$\sigma^{(1)} = -15319,7$$

$$\sigma^{(2)} = 3133,4$$

$$\sigma^{(3)} = 20186,3.$$

(e) For the maximum principal stress $\lambda_3 = 20186,3$ we solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

I.e.

$$\begin{pmatrix} 20000 - 20186,3 & 2000 & 1000 \\ 2000 & -15000 - 20186,3 & 2000 \\ 1000 & 2000 & 3000 - 20186,3 \end{pmatrix} \cdot \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Solving this gives $n_1 = 15,3299$, $n_2 = 0,928198$, $n_3 = 1$. Now we scale this by the length to get a unit vector:

$$\mathbf{n}_{\max} = \frac{\begin{pmatrix} 15,3299 \\ 0,928198 \\ 1 \end{pmatrix}}{\sqrt{15,3299^2 + 0,928198^2 + 1^2}} = \begin{pmatrix} 0,996 \\ 0,0603 \\ 0,0650 \end{pmatrix}.$$

(a) $\mathbf{T} = \frac{1}{\sqrt{3}} \begin{pmatrix} 23000 \\ -11000 \\ 6000 \end{pmatrix}$

(b) $\mathbf{N} = \frac{1}{\sqrt{3}} \begin{pmatrix} 6000 \\ 6000 \\ 6000 \end{pmatrix}$

$\mathbf{S} = \frac{1}{\sqrt{3}} \begin{pmatrix} 17000 \\ -17000 \\ 0 \end{pmatrix}$

(d) $\sigma^{(1)} = -15319,7$
 $\sigma^{(2)} = 3133,4$
 $\sigma^{(3)} = 20186,3$

(e) $\mathbf{n}_{\max} = \begin{pmatrix} 0,996 \\ 0,060 \\ 0,065 \end{pmatrix}.$

RESULT

Problem 2.3

The state of stress at a point is given by the components of stress tensor σ_{ij} . A plane is defined by the direction cosines of the normal $(1/2, 1/2, 1/\sqrt{2})$. State the general conditions for which the traction on the plane has the same direction as the x_2 -axis and a magnitude of 1,0.

Derivation

We use

$$T_i = \sigma_{ji} n_j = \boldsymbol{\sigma} \cdot \mathbf{n}.$$

We are given

$$\mathbf{n} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{pmatrix}.$$

The traction should be in the x_2 axis and have magnitude 1,0 so it must be

$$\mathbf{T} = \begin{pmatrix} 0 \\ 1,0 \\ 0 \end{pmatrix} = \mathbf{e}_2.$$

So the condition is simply

$$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2.$$

$\boldsymbol{\sigma} \mathbf{n} = \mathbf{e}_2$

RESULT

Problem 2.4

Determine the body forces for which the following stress field describes a state of equilibrium

$$\sigma_{xx} = -2x^2 - 3y^2 - 5z,$$

$$\sigma_{xy} = x + 4xy - 6,$$

$$\sigma_{yy} = -2y^2 + 7,$$

$$\sigma_{xz} = -3x + 2y + 1.$$

$$\sigma_{zz} = 4x + y + 3z - 5,$$

$$\sigma_{yz} = 0.$$

Derivation

We have that

$$\sigma_{ij,j} + f_i = 0.$$

which is:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} + f_x = 0$$

$$\frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} + f_y = 0$$

$$\frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + f_z = 0.$$

In the x -direction we get

$$\begin{aligned} \frac{\partial(-2x^2 - 3y^2 - 5z)}{\partial x} + \frac{\partial(x + 4xy - 6)}{\partial y} + \frac{\partial(-3x + 2y + 1)}{\partial z} + f_x &= 0 \\ -4x + 4x + f_x &= 0 \\ f_x &= 0. \end{aligned}$$

In the y -direction we get

$$\begin{aligned} \frac{\partial(x + 4xy - 6)}{\partial x} + \frac{\partial(-2y^2 + 7)}{\partial y} + \frac{\partial 0}{\partial z} + f_y &= 0 \\ 1 + 4y - 4y + f_y &= 0 \\ f_y &= -1. \end{aligned}$$

And in the z -direction we get

$$\begin{aligned} \frac{\partial(-3x + 2y + 1)}{\partial x} + \frac{\partial 0}{\partial y} + \frac{\partial(4x + y + 3z - 5)}{\partial z} + f_z &= 0 \\ -3 + 3 + f_z &= 0 \\ f_z &= 0. \end{aligned}$$

$$\mathbf{f} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} = -\mathbf{e}_2$$

RESULT

Problem 2.5

Determine whether the following stress field is admissible in an elastic body when body forces are negligible.

$$\boldsymbol{\sigma} = \begin{bmatrix} yz+4 & z^2+2x & 5y+z \\ \cdot & xz+3y & 8x^3 \\ \cdot & \cdot & 2xyz \end{bmatrix}.$$

Derivation

We once again use

$$\sigma_{ij,j} + f_i = 0.$$

With negligible body forces this becomes

$$\sigma_{ij,j} = 0.$$

I.e.

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} &= 0 \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} &= 0 \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} &= 0. \end{aligned}$$

In the x -direction we get

$$\begin{aligned} \frac{\partial(yz+4)}{\partial x} + \frac{\partial(z^2+2x)}{\partial y} + \frac{\partial(5y+z)}{\partial z} &= 0 \\ 0 + 0 + 1 &\neq 0. \end{aligned}$$

Hence, the stress field is not admissible with negligible body forces.

It is not.

RESULT

Problem 2.8

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}.$$

- Show that the principal stresses are 4, 1, and -2 .
- Find the orientation of the principal planes

Derivation

- We have the characteristic equation

$$|\sigma_{ji} - \lambda \delta_{ij}| = 0.$$

I.e.

$$\begin{vmatrix} 3-\lambda & 1 & 1 \\ 1 & 0-\lambda & 2 \\ 1 & 2 & 0-\lambda \end{vmatrix} = 0 \Rightarrow \lambda_1 \rightarrow -2, \quad \lambda_2 \rightarrow 1, \quad \lambda_3 \rightarrow 4.$$

- The orientation of the principal planes are the eigenvectors of these. I.e. we have to solve

$$(\sigma_{ji} - \lambda \delta_{ij}) n_j = 0.$$

For each λ .

For $\lambda_1 = -2$ we get

$$\begin{bmatrix} 5 & 1 & 1 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{bmatrix} \cdot \mathbf{n}_1 = 0 \Rightarrow \mathbf{n}_1 = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}.$$

For $\lambda_2 = 1$ we get

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 2 & -1 \end{bmatrix} \mathbf{n}_2 = 0 \Rightarrow \mathbf{n}_2 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}.$$

For $\lambda_3 = 4$ we get

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -4 & 2 \\ 1 & 2 & -4 \end{bmatrix} \mathbf{n}_3 = 0 \Rightarrow \mathbf{n}_3 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

$$\hat{\mathbf{n}}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

$$(a) \quad \sigma^{(1)} = -2$$

$$\sigma^{(2)} = 1$$

$$\sigma^{(3)} = 4$$

$$(b) \quad \hat{n}_1 = \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$\hat{n}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$\hat{n}_3 = \begin{pmatrix} \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \end{pmatrix}.$$

RESULT

Problem 2.9

The stress tensor at a point $P(x, y, z)$ in a solid is

$$\boldsymbol{\sigma} = \begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix}.$$

- (a) Determine the stresses with respect to a set of axes x' , y' , and z' for which the matrix of direction cosines is

$$[\alpha_{ij}] = \begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

- (b) Compute the principal stresses using both the x -, y -, and z - and x' -, y' -, and z' -axes as the basis.
 (c) Find the orientation of the principal planes with respect to the x -, y -, and z -axes.

Derivation

- (a) To transform coordinates we use

$$\mathbf{A}' = \mathbf{R} \mathbf{A} \mathbf{A}^T.$$

Hence

$$\boldsymbol{\sigma}' = \boldsymbol{\alpha} \boldsymbol{\sigma} \boldsymbol{\alpha}^T$$

which gives

$$\boldsymbol{\sigma}' = \begin{pmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{bmatrix}.$$

- (b) Here we use the characteristic equation from before to find the eigenvalues:

$$\begin{bmatrix} 120 - \lambda & 40 & 30 \\ 40 & 150 - \lambda & 20 \\ 30 & 20 & 100 - \lambda \end{bmatrix} = 0 \implies \lambda_1 = 190,405, \quad \lambda_2 = 102,423, \quad \lambda_3 = 77,1724.$$

and

$$\begin{bmatrix} 107,86 - \lambda & -32,99 & -2,32 \\ -32,99 & 162,14 - \lambda & 35,98 \\ -2,32 & 35,98 & 100 - \lambda \end{bmatrix} \rightarrow \lambda_1 = 190,403, \quad \lambda_2 = 102,424, \quad \lambda_3 = 77,1727.$$

Which confirms that principal stresses are unchanged by rotation.

- (c) Here we solve

$$|\sigma_{ji} - \lambda \delta_{ij}| n_j = 0$$

for each eigenvalue and get:

$$\begin{bmatrix} 120 & 40 & 30 \\ 40 & 150 & 20 \\ 30 & 20 & 100 \end{bmatrix} \rightarrow \left(\begin{array}{ccc} 190.405 & 102.423 & 77.1724 \\ \{1.61658, 2.09537, 1.\} & \{0.821389, -1.11094, 1.\} & \{-0.911651, 0.226096, 1.\} \end{array} \right).$$

Now we scale each $\mathbf{n}$ and get:

$$\mathbf{n}_1 = \begin{pmatrix} 0,5714 \\ 0,7406 \\ 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 \\ -0,6514 \\ 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 \\ -0,1648 \\ -0,7289 \end{pmatrix}.$$

$$(a) \quad \boldsymbol{\sigma} = \begin{pmatrix} 107,86 & -32,99 & -2,32 \\ -32,99 & 162,14 & 35,98 \\ -2,32 & 35,98 & 100 \end{pmatrix}$$

$$(b) \quad \lambda_1 = 190,405$$

$$\lambda_2 = 102,423$$

$$\lambda_3 = 77,1724$$

$$(c) \quad \mathbf{n}_1 = \begin{pmatrix} 0,5714 & 0,7406 & 0,3535 \end{pmatrix}$$

$$\mathbf{n}_2 = \begin{pmatrix} 0,4816 & -0,6514 & 0,5863 \end{pmatrix}$$

$$\mathbf{n}_3 = \begin{pmatrix} 0,6645 & -0,1648 & -0,7289 \end{pmatrix}.$$

RESULT

Lecture 5: The Strain Tensor

23. Februar 2026

Problem 3.1

The components of a displacement field are

$$\begin{aligned}u_x &= (x^2 + 20) \cdot 10^{-4}, \\u_y &= 2yz \cdot 10^{-3}, \\u_z &= (z^2 - xy) \cdot 10^{-3}.\end{aligned}$$

- (a) Consider two points in the undeformed system, $(2, 5, 7)$ and $(3, 8, 9)$. Find the change in distance between these points.

Derivation

The displacement at $A = (2, 5, 7)$ is

$$\begin{aligned}u_x(A) &= (2^2 + 20) \cdot 10^{-4} = 0,0024 \\u_y(A) &= 2 \cdot 5 \cdot 7 \cdot 10^{-3} = 0,070 \\u_z(A) &= (7^2 - 2 \cdot 5) \cdot 10^{-3} = 0,039\end{aligned}$$

so

$$\mathbf{u}(A) = \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix}.$$

And at $B = (3, 8, 9)$ it is

$$\begin{aligned}u_x(B) &= (3^2 + 20) \cdot 10^{-4} = 0,0029 \\u_y(B) &= 2 \cdot 8 \cdot 9 \cdot 10^{-3} = 0,144 \\u_z(B) &= (9^2 - 3 \cdot 8) \cdot 10^{-3} = 0,057\end{aligned}$$

so

$$\mathbf{u}(B) = \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix}.$$

The deformed positions are:

$$\begin{aligned}A' &= A + \mathbf{u}(A) = \begin{pmatrix} 2 \\ 5 \\ 7 \end{pmatrix} + \begin{pmatrix} 0,0024 \\ 0,070 \\ 0,039 \end{pmatrix} = \begin{pmatrix} 2,0024 \\ 5,070 \\ 7,039 \end{pmatrix} \\B' &= B + \mathbf{u}(B) = \begin{pmatrix} 3 \\ 8 \\ 9 \end{pmatrix} + \begin{pmatrix} 0,0029 \\ 0,144 \\ 0,057 \end{pmatrix} = \begin{pmatrix} 3,0029 \\ 8,144 \\ 9,057 \end{pmatrix}.\end{aligned}$$

Our position vectors are:

$$AB = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$$

$$AB' = \begin{pmatrix} 1.0005 \\ 3.074 \\ 2.018 \end{pmatrix}.$$

These have the lengths:

$$|AB| = \sqrt{1^2 + 3^2 + 2^2} = 3.74166$$

$$|AB'| = \sqrt{1.0005^2 + 3.074^2 + 2.018^2} = 3.81088.$$

The change in length is then:

$$\Delta L = |AB'| - |AB| = 3.81088 - 3.74166 = 0.06922.$$

$\Delta L = 0.06922$

RESULT

Lecture 6: Physical interpretation of the strain tensor

2. Marts 2026

Problem 3.2

The strain field at a point $P(x, y, z)$ in an elastic body is given by

$$\epsilon = \begin{bmatrix} 20 & 3 & 2 \\ 3 & -10 & 5 \\ 2 & 5 & -8 \end{bmatrix} \times 10^{-6}.$$

Determine the following values:

- (a) The strain invariants
- (b) The principal strains
- (e) The mean normal strain and the strain deviator

Derivation

- (a) The strain invariants are Q_1, Q_2, Q_3 , which is the sum of the diagonal of ϵ , the sum of the minors of the diagonal of ϵ and the determinant of ϵ , respectively. The first strain invariant Q_1 is simply the sum of the diagonal entries of ϵ as

$$Q_1 = 20 \cdot 10^{-6} + (-10 \cdot 10^{-6}) + (-8 \cdot 10^{-6}) = 2 \cdot 10^{-6}.$$

The second strain invariant Q_2 is the sum of the minors of the diagonal, as:

$$Q_2 = \begin{vmatrix} 20 \cdot 10^{-6} & 3 \cdot 10^{-6} \\ 3 \cdot 10^{-6} & -10 \cdot 10^{-6} \end{vmatrix} + \begin{vmatrix} 20 \cdot 10^{-6} & 2 \cdot 10^{-6} \\ 2 \cdot 10^{-6} & -8 \cdot 10^{-6} \end{vmatrix} + \begin{vmatrix} -10 \cdot 10^{-6} & 5 \cdot 10^{-6} \\ 5 \cdot 10^{-6} & -8 \cdot 10^{-6} \end{vmatrix} = -3,18 \cdot 10^{-10}.$$

The third strain invariant is the determinant of ϵ , as

$$Q_3 = \begin{vmatrix} 20 \cdot 10^{-6} & 3 \cdot 10^{-6} & 2 \cdot 10^{-6} \\ 3 \cdot 10^{-6} & -10 \cdot 10^{-6} & 5 \cdot 10^{-6} \\ 2 \cdot 10^{-6} & 5 \cdot 10^{-6} & -8 \cdot 10^{-6} \end{vmatrix} = 1,272 \cdot 10^{-15}.$$

- (b) The principal strains are found by solving the eigenvalue problem:

$$\begin{bmatrix} 20 & 3 & 2 \\ 3 & -10 & 5 \\ 2 & 5 & -8 \end{bmatrix} \rightarrow \{20.519, -14.1326, -4.38641\}.$$

- (c) The mean normal strain is simply the mean of the three normal, i.e. ϵ_{ii} components of ϵ :

$$\epsilon_M = \frac{1}{3} (\epsilon_{11} + \epsilon_{22} + \epsilon_{33}) = \frac{1}{3} \cdot (20 + -10 + -8) \cdot 10^{-6} = 0,666 \cdot 10^{-6}.$$

Which means the mean normal strain tensor is:

$$\epsilon_M = \begin{bmatrix} 0,666 \cdot 10^{-6} & 0 & 0 \\ 0 & 0,666 \cdot 10^{-6} & 0 \\ 0 & 0 & 0,666 \cdot 10^{-6} \end{bmatrix}.$$

The strain deviator tensor is then:

$$\epsilon_D = \epsilon - \epsilon_M = \begin{bmatrix} 20 - 0,666 & 3 & 2 \\ 3 & -10 - 0,666 & 5 \\ 2 & 5 & -8 - 0,666 \end{bmatrix} 10^{-6} = \begin{pmatrix} 19.334 & 3. & 2. \\ 3. & -10.666 & 5. \\ 2. & 5. & -8.666 \end{pmatrix} 10^{-6}.$$

$$\begin{aligned}
 Q_1 &= 2 \cdot 10^{-6} \\
 \text{(a)} \quad Q_2 &= -3,18 \cdot 10^{-10} \\
 Q_3 &= 1,272 \cdot 10^{-15} \\
 \epsilon^{(1)} &= 20,519 \\
 \text{(b)} \quad \epsilon^{(2)} &= -14,132 \\
 \epsilon^{(3)} &= -4,386 \\
 \epsilon_M &= \begin{bmatrix} 0,666 \cdot 10^{-6} & 0 & 0 \\ 0 & 0,666 \cdot 10^{-6} & 0 \\ 0 & 0 & 0,666 \cdot 10^{-6} \end{bmatrix} \\
 \text{(c)} \quad \epsilon_D &= \begin{bmatrix} 19,334 \cdot 10^{-6} & 3 \cdot 10^{-6} & 2 \cdot 10^{-6} \\ 3 \cdot 10^{-6} & -10,666 \cdot 10^{-6} & 5 \cdot 10^{-6} \\ 2 \cdot 10^{-6} & 5 \cdot 10^{-6} & -8,666 \cdot 10^{-6} \end{bmatrix}.
 \end{aligned}$$

RESULT

Problem 3.3

With reference to Problem 3.2, consider a second coordinate system x'_i in which x'_2 is parallel to x_2 and x'_1 and x'_3 are defined by a 60° counterclockwise rotation about x_2 . Find the components of strain with reference to the x'_i system using the strain field from Problem 3.2.

Derivation

We use the tensor transformation rule

$$\mathbf{A}' = \mathbf{R} \mathbf{A} \mathbf{R}^T$$

where $\mathbf{R}$ is the direction cosine matrix whose rows are the new basis vectors written in the old basis. In this case the tensor ϵ is to be rotated around the x_2 -axis, which means:

$$\mathbf{R} = \begin{bmatrix} \cos 60^\circ & 0 & \sin 60^\circ \\ 0 & 1 & 0 \\ -\sin 60^\circ & 0 & \cos 60^\circ \end{bmatrix}.$$

So

$$\epsilon' = \begin{bmatrix} \cos 60^\circ & 0 & \sin 60^\circ \\ 0 & 1 & 0 \\ -\sin 60^\circ & 0 & \cos 60^\circ \end{bmatrix} \cdot \begin{bmatrix} 20 & 3 & 2 \\ 3 & -10 & 5 \\ 2 & 5 & -8 \end{bmatrix} \cdot \begin{bmatrix} \cos 60^\circ & 0 & \sin 60^\circ \\ 0 & 1 & 0 \\ -\sin 60^\circ & 0 & \cos 60^\circ \end{bmatrix}^T = \begin{pmatrix} 5,86603 & 0 & 5,13397 \\ 0 & -10 & 0 \\ -14,134 & 0 & -2,86603 \end{pmatrix}.$$

$$\begin{bmatrix} 5,866 & 0 & 5,134 \\ 0 & -10 & 0 \\ -14,134 & 0 & -2,866 \end{bmatrix}$$

RESULT

Problem 3.6

Compute the infinitesimal strains, and sketch the deformed configuration of an initially rectangular volume element subject to the following displacement fields, where A , B , and D_{ij} = constants:

- (a) Simple extension: $u_1 = A_1 x_1; u_2 = u_3 = 0$.
- (b) Simple shear: $u_1 = B x_2; u_2 = u_3 = 0$.
- (c) homogeneous deformation: $u_i = D_{ij} x_j$.

Derivation

- (a) Here $u_1 = A_1 x_1$ and $u_2 = u_3 = 0$ leading to the following displacement gradient

$$\nabla \mathbf{u} = \begin{bmatrix} A_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Here, the block simply stretches/contracts in the x_1 -direction.

- (b) Here $u_1 = B x_2$ and $u_2 = u_3 = 0$ leading to the following displacement gradient

$$\nabla \mathbf{u} = \begin{bmatrix} 0 & B & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Which is symmetrized as:

$$\epsilon = \frac{1}{2} (\nabla \mathbf{u} + (\nabla \mathbf{u})^T) = \frac{1}{2} \begin{bmatrix} 0 & \frac{B}{2} & 0 \\ \frac{B}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

here we have a displacement in the x_1 direction proportional to x_2 .

- (c) Here we have the general case with deformation in all directions. The displacement is

$$\epsilon = \frac{1}{2} (\mathbf{D} + (\mathbf{D})^T).$$

$$\begin{aligned} \text{(a)} \quad \epsilon &= \begin{bmatrix} A_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \text{(b)} \quad \epsilon &= \frac{1}{2} \begin{bmatrix} 0 & \frac{B}{2} & 0 \\ \frac{B}{2} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \text{(c)} \quad \epsilon &= \frac{1}{2} (\mathbf{D} + (\mathbf{D})^T). \end{aligned}$$

RESULT

Lecture 7: Volume and shape change

9. Marts 2026

Lecture 8: Linear isotropic materials and the uniaxial stress

16. Marts 2026

Problem 4.2

For an isotropic material, obtain the relationship

$$\epsilon_{ij} = \frac{1}{2\mu} \sigma_{ij} - \frac{\lambda}{2\mu(2\mu + 3\lambda)} \delta_{ij} \sigma_{kk}$$

from the expression

$$\sigma_{ij} = 2\mu\epsilon_{ij} + \lambda\delta_{ij}\epsilon_{kk}.$$

Problem 4.3a

Show that the principal axes of stress correspond with the principal axes of strain for an isotropic, linearly elastic material.

Problem 4.5

For steel, $E = 30 \cdot 10^6$ and $G = 12 \cdot 10^6$ (force/length²). The components of strain at a point within this material are given by

$$\epsilon = \begin{bmatrix} 0,004 & 0,001 & 0 \\ 0,001 & 0,006 & 0,004 \\ 0 & 0,004 & 0,001 \end{bmatrix}.$$

Compute the corresponding components of the stress tensor.

Lecture 9: Thermal strains

23. Marts 2026

Problem 4.9

Consider an isotropic elastic medium subject to a hydrostatic state of stress:

$$\sigma^{(1)} = \sigma^{(2)} = \sigma^{(3)} = -p.$$

Show that for this state of stress

$$p = -KR_1,$$

where K is the bulk modulus and R_1 is the cubical strain. Consider the implications of $\nu > 1/2$ and $\nu < -1$.

Problem 4.10

For an elastic medium subject to a state of stress σ_{ij} , assume that the deformation is incompressible for $\sigma_{(ii)} \neq 0$.

- (a) Verify that $\nu = 0,5$.
- (b) Assume also that $\epsilon_{33} = 0$ and determine ν .

Problem 5.1a

The stress components at a point $P(x, y, z)$ in an isotropic medium are

$$\begin{aligned}\sigma_{xx} &= y + 3z^2, & \sigma_{xy} &= z^3, \\ \sigma_{yy} &= x + 2z, & \sigma_{xz} &= y^2, \\ \sigma_{zz} &= 2x + y, & \sigma_{yz} &= x^2.\end{aligned}$$

No body forces are present.

Does this stress field satisfy equilibrium?

Problem 5.3

For the following stress field,

$$\begin{aligned}\sigma_{xx} &= 80x^3 + y, & \sigma_{xy} &= 1000 + 100y^2, \\ \sigma_{yy} &= 100x^3 + 1600, & \sigma_{xz} &= 0, \\ \sigma_{zz} &= 90y^2 + 100z^3, & \sigma_{yz} &= xz^3 + 100x^2y.\end{aligned}$$

- (a) What is the body force distribution required for equilibrium?
- (c) With $E = 30 \cdot 10^6$ (force/length²) and $\nu = 0,3$, what is the strain at $(2, 2, 5)$?